TOPSTECHNOLOGIES
Training | Outsourcing | Placement | Study Abroad

Name: vrajesh

 # Network Threats and Mitigation & Physical Security and Risk

1. Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

Answer:

| Network Threat | Description | Real-World Example |

| ------------------------------------ | -- | -- |

*| Phishing | Fake emails, messages, or websites that trick users into revealing passwords or banking details. | A fake **Paytm** SMS asking users to click a link and enter their login credentials. |*

*| DDoS (Distributed Denial of Service) | Attackers flood a server with huge amounts of traffic to make it unavailable. | An **Instagram** server experiencing heavy fake traffic, causing users to be unable to log in temporarily. |*

*| Man-in-the-Middle (MITM) | An attacker secretly intercepts communication between two users or devices. | Using **WhatsApp** on an unsafe public Wi-Fi where an attacker tries to intercept network traffic. |*

| Malware | Harmful software that steals data, damages files, or controls a device.

*| Downloading a fake **WhatsApp APK** that installs malware on a phone. |*

*| Password Attack | Attackers try many passwords to gain access to an account. | Someone repeatedly tries different passwords to log into an **Instagram** account. |*

Explanation:

These threats target user accounts, personal information, and online services. Companies use encryption, firewalls, authentication, and monitoring systems to protect users from these attacks.

2. Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN

code by checking all possible combinations against a preset value.

Constraint: The script should print the number of attempts it took to guess the correct PIN.

Answer:

Python Code

```python
# Brute Force PIN Simulation

correct_pin = "4829"
attempts = 0
```

```
for i in range(10000):
    guess = str(i).zfill(4)
    attempts += 1

    if guess == correct_pin:
        print("PIN Found:", guess)
        print("Attempts:", attempts)
        break
```

Sample Output

```
PIN Found: 4829
Attempts: 4830
```

Explanation:

* The program checks every possible PIN from **0000** to **9999**.

* It compares each PIN with the correct PIN.

* When the correct PIN is found, it prints the PIN and the total number of attempts.

---

3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

Answer:

1. Unauthorized Access

Risk: An unauthorized person may enter the server room and access important servers or networking devices.

Mitigation: Install biometric authentication or keycard access so that only authorized staff can enter.

2. Fire

Risk: A fire can damage servers, routers, switches, and important company data.

Mitigation: Install smoke detectors, fire alarms, and an automatic fire suppression system.

3. Power Failure

Risk: A sudden power outage can shut down servers and may cause data loss or service interruption.

Mitigation: Use a UPS (Uninterruptible Power Supply) and a backup generator to provide continuous power.

4. Theft of Equipment

Risk: Servers, routers, switches, or storage devices may be stolen from the server room.

Mitigation: Install CCTV cameras, lock the server racks, and keep the server room secured at all times.

Explanation:

Physical security protects servers and networking equipment from damage, theft, or unauthorized access. Proper security measures help ensure that online services continue to operate safely.

---

4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).

Answer:

Office Floor Plan

```text
+---------------------------------------------+
|               OFFICE ENTRANCE               |
|        [Biometric Access Door]              |
```

```
        |                ||                 |
        |                ||                 |
|   CCTV Camera   +------------------------+   |
      |      📷        |                  |   |
   |             |    Employee Area      |   |
   |             |  PC1     PC2     PC3  |   |
      |             |   |     |      | |   |
   |             | Cable    Cable    Cable|   |
   |             | Locks    Locks    Locks|   |
   |             +------------------------+   |
        |                                   |
   |      +----------------------------+      |
   |      |       SERVER ROOM          |      |
   |      |  Router   Switch   Server  |      |
        |      |                     |      |
  |      | 🔓 Locked Server Rack      |      |
```

| | 📷 CCTV Camera | |

| +----------------------------+ |

+--+

```

### Physical Security Measures Used

1. **Biometric access** at the entrance to allow only authorized people.
2. **CCTV cameras** to monitor the office and server room.
3. **Cable locks and locked server racks** to prevent theft of computers and networking equipment.

---
```

5. Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

Answer:

College Computer Lab Security Checklist

1. Install and regularly update antivirus software on every computer.

2. Enable Windows Firewall on all lab computers.

3. Use strong passwords and change administrator passwords regularly.

4. Lock the server room with biometric or keycard access.

5. Install CCTV cameras to monitor the computer lab and server room.

6. Regularly update the operating system and all software to fix security vulnerabilities.

7. Back up important student records and lab data to a secure location on a regular schedule.

Explanation:

*This checklist combines both **network security** and **physical security** practices. Following these steps helps protect the college computer lab from cyber attacks, unauthorized access, hardware theft, and data loss.*